

Unit IV

4.4. Heterogeneous Catalysis

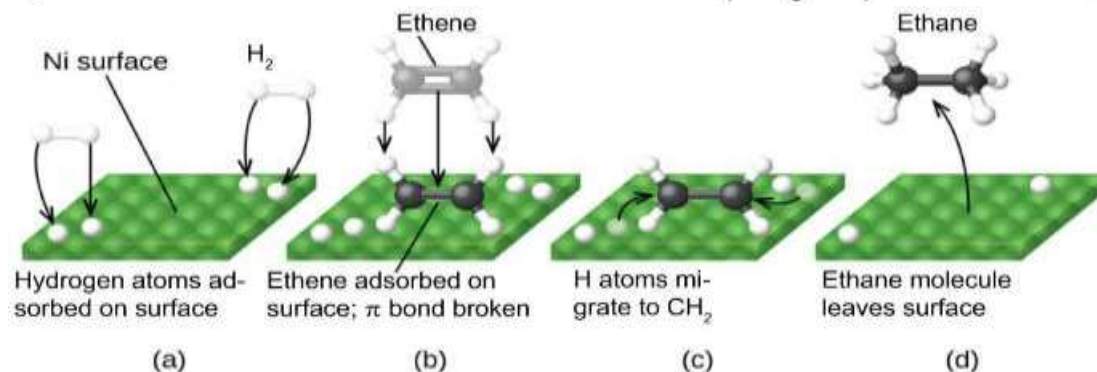
The catalytic process in which the reactants and the catalyst are in different phases is known as heterogeneous catalysis. such catalysts are usually solids, and often function by furnishing an active surface upon which one or more steps in the reaction can occur.

- In heterogeneous catalysis, the molecules react at the surface of a solid catalyst
- The mode of action of a heterogeneous catalyst consists of the following steps:
- **Adsorption** (or chemisorption) of the reactants on the catalyst surface
 - The reactants diffuse to the surface of the catalyst
 - The reactant is physically adsorbed onto the surface by weak forces
 - The reactant is chemically adsorbed onto the surface by stronger bonds
 - **Chemisorption** causes bond weakening between the atoms of the reactants
- **Desorption** of the products
 - The bonds between the products and catalyst weaken so much that the products break away from the surface
 - For example, the adsorption of hydrogen molecules onto a palladium (Pd) surface

4.4.1. Contact theory (or) mechanism of heterogeneous catalysis (or) adsorption theory

This theory postulated by Faraday in 1883. It explains the action of heterogeneous catalysis. Heterogeneous catalysis has five steps.

1. **Diffusion of Reactant(s) to the Surface:** The reactants diffuse to the surface of the catalysts. Some of the reactant cross the barrier and enter the interior exposed surface that includes paths and cracks on the external surface.
2. **Adsorption of reactants:** Bonds are formed as the reactant(s) are adsorbed onto the surface of the catalyst. The ability for an atom or molecule to stick to the surface is known as the Sticking Co-efficient.
3. **Reaction:** The reactants, when bound to the surface have a higher probability of reacting with each other, and after the reaction, they form an intermediate compound.
4. **Desorption of products:** The intermediate compound gets desorbed from the surface, which again becomes available for adsorption for other molecules
5. **Diffusion of Product(s):** The intermediate compound then disintegrates to form the final products and the products are then desorbed from the surface of the catalyst.

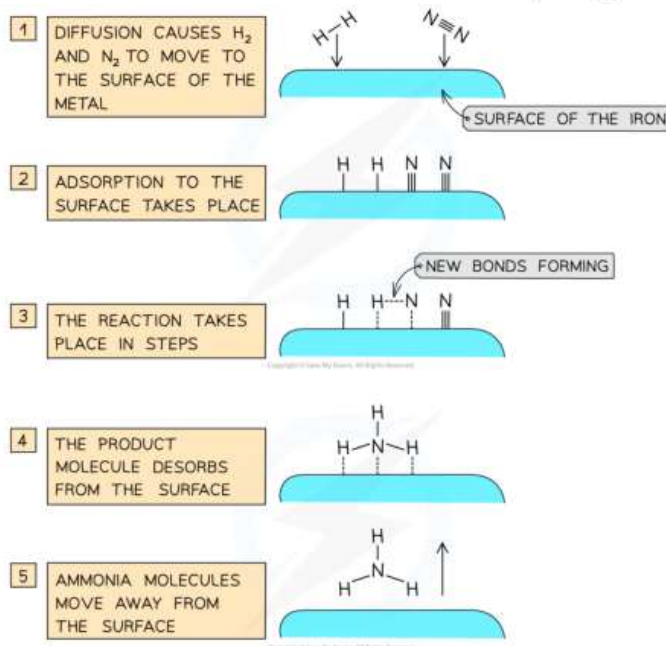


There are four steps in the catalytic hydrogenation of ethene on a nickel surface, $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$. (a) Hydrogen is adsorbed on the surface, breaking the H-H bonds and forming Ni-H bonds. (b) Ethene is adsorbed on the surface, breaking the π bond and forming Ni-C bonds. (c) H atoms diffuse across the surface and form new C-H bonds when they reach ethene molecules. (d) The saturated carbon atoms in C_2H_6 molecules can no longer bond to the surface so the ethane molecules escape from the surface.

4.4.2. Examples of Heterogeneous Catalysis

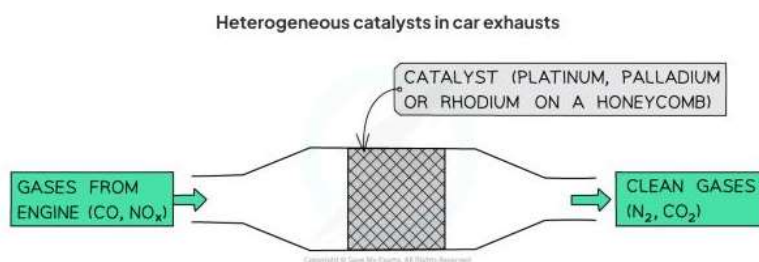
Iron in the Haber process

- In the Haber process ammonia (NH_3) is produced from nitrogen (N_2) and hydrogen (H_2)
- An iron catalyst is used which speeds up the reaction by bringing the reactants close together on the metal surface
- This increases their likelihood to react with each other
- The mode of action of the iron catalyst is as follows:
 - Diffusion of the nitrogen and hydrogen gas to the iron surface
 - Adsorption of the reactant molecules onto the iron surface by forming bonds between the iron and reactant atoms
 - These bonds are so strong that they weaken the covalent bonds between the nitrogen atom in N_2 and hydrogen atoms in H_2
 - But they are weak enough to break when the catalysis has been completed
- The reaction takes place between the adsorbed nitrogen and hydrogen atoms which react with each other on the iron surface to form NH_3
- Desorption occurs when the bonds between the NH_3 and iron surface are weakened and
- eventually broken
- The formed NH_3 diffuses away from the iron surface



Heterogeneous catalysts in catalytic converters

- Heterogeneous catalysts are also used in the catalytic removal of oxides of nitrogen from the exhaust gases of car engines
- The catalysts speed up the conversion of:
 - Nitrogen oxides (NO_x) into harmless nitrogen gas (N_2)
 - Carbon monoxide (CO) into carbon dioxide (CO_2)
- The catalytic converter has a honeycomb structure containing small beads coated with platinum, palladium, or rhodium metals which act as heterogeneous catalysts
- The mode of action of the catalysts is as follows:
 - Adsorption of the nitrogen oxides and CO onto the catalyst surface
 - The weakening of the covalent bonds within nitrogen oxides and CO
 - Formation of new bonds between:
 - Adjacent nitrogen atoms to form N_2 molecules
 - CO and oxygen atoms to form CO_2 molecules
- Desorption of N_2 and CO_2 molecules which eventually diffuse away from the metal surface

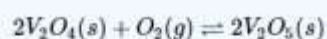
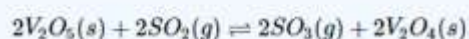
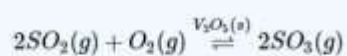


The metals in catalytic converters speed up the conversion of nitrogen oxides and CO into N_2 and CO_2 respectively

Contact Process

- The contact process is the current method of producing sulfuric acid in the high concentrations needed for industrial processes.
- Platinum used to be the catalyst for this reaction; however, as it is susceptible to reacting with arsenic impurities in the sulfur feedstock, vanadium(V) oxide (V_2O_5) is now preferred.
- In the contact process vanadium [V] oxide (V_2O_5) is solid whereas the reactants SO_2 and O_2 are gaseous.

More detailed method:



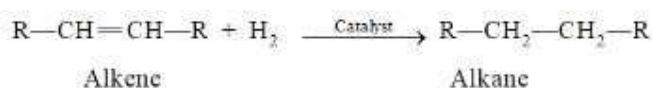
Therefore:



Catalytic hydrogenation

- In this process the liquid Unsaturated hydrocarbons (alkenes) reacts with gaseous hydrogen (H_2) to produce liquid saturated hydrocarbons (alkanes) where metals like platinum (Pt) and palladium (Pd) as the catalyst.

Conversion of ethylene to ethane



Catalyst :

- Pd/Pt at ordinary temp. and pressure
- Ni, 200–300° C (sabatier)
- Raney Ni at room temp.

Process	Catalyst	Reaction
Making ammonia	Iron	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
Making synthesis gas (carbon monoxide and hydrogen)	Nickel	$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$
Catalytic cracking of gas oil	Zeolite	Produces: a gas (e.g. ethene, propene) a liquid (e.g. petrol) a residue (e.g. fuel oil)
Reforming of naphtha	Platinum and rhenium on alumina	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3(\text{g}) \rightarrow \text{C}_6\text{H}_{12}(\text{g}) + \text{H}_2(\text{g})$
Making epoxyethane	Silver on alumina	$\text{C}_2\text{H}_4(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{C}-\text{O}-\text{CH}_2(\text{g})$
Making sulfuric acid	Vanadium(V) oxide on silica	$\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{SO}_3(\text{g})$
Making nitric acid	Platinum and rhodium	$4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$

4.4.3. Key Challenges of Heterogeneous Catalysis in Industrial Practices

Deactivation and Poisoning

One of the most significant challenges in heterogeneous catalysis is the deactivation of the catalyst over time. This can occur due to:

- **Chemical Impurities:** Impurities in the reactants can accumulate on the catalyst's surface, blocking active sites and reducing its effectiveness.
- **Reaction By-products:** Certain heterogeneous catalytic reactions can produce by-products that coat the catalyst, similarly inhibiting its activity.

These factors lead to a gradual loss of catalyst performance, necessitating frequent regeneration or replacement, which can be costly and time-consuming.

Mass Transfer Limitations

The efficiency of a heterogeneous catalyst is heavily dependent on the ability of reactants to reach and interact with active sites on the catalyst surface. Challenges include:



- **Diffusion Barriers:** Poor diffusion of reactants to the catalyst surface can limit reaction rates, particularly in catalysts with low porosity or improper pore size distribution.
- **Product Removal:** Inefficient removal of products from the catalyst surface can also hinder the overall reaction rate, as it prevents fresh reactants from accessing active sites.

Optimizing the physical structure of the catalyst to enhance mass transfer is crucial for maximizing its performance.

Thermal Stability

Heterogeneous catalysts are often used under harsh conditions, including high temperatures. However, not all catalysts can withstand such environments:

- **Structural Degradation:** High temperatures can cause structural changes in the catalyst, such as sintering, where particles coalesce and reduce the surface area available for reaction.
- **Activity Loss:** Thermal instability can lead to a decrease in catalytic activity, compromising the process efficiency and leading to potential process failures.

Developing catalysts with enhanced thermal stability is essential for applications involving high temperatures.

Scalability

Translating catalytic processes from the laboratory to industrial scale presents several challenges:

- **Replicating Conditions:** Conditions optimized in small-scale experiments often do not translate directly to larger systems due to differences in mixing, heat transfer, and mass transfer.
- **Cost and Material Availability:** Scaling up often requires significantly more catalyst material, which can be expensive and logistically challenging if rare or costly materials are involved.

4.4.4. Difference Between Homogeneous and Heterogeneous Catalyst

The key difference between homogeneous and heterogeneous catalysts is that homogeneous catalysts can always be found in the liquid phase whereas heterogeneous catalysts can be found

in all three phases of matter; solid phase, liquid phase, and gas phase.



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Homogeneous vs Heterogeneous Catalyst	
Homogeneous catalysts are catalytic compounds that are in the same phase as the substances which are going into the reaction phase.	Heterogeneous catalysts are catalytic compounds that are in a different phase from that of the phase of the reaction mixture.
Phase	
Homogeneous catalysts can be found mostly in the liquid phase.	Heterogeneous catalysts can be found in all three phases; solid phase, liquid phase or gas phase.
Thermal Stability	
The thermal stability of homogeneous catalysts is poor.	The thermal stability of heterogeneous catalysts is good.
Catalyst Recovery	
The recovery of homogeneous catalysts is difficult and expensive.	The recovery of heterogeneous catalysts is easy and cheap.
Active Site	
The active site of homogeneous catalysts are well-defined and has good selectivity.	The active site of heterogeneous catalysts are not well-defined and has poor selectivity.
Catalyst Recycling	
The recycling of homogeneous catalysts is difficult.	The recycling of heterogeneous catalysts is easy.
Catalyst Separation	
The separation of homogeneous catalyst from the reaction mixture is difficult.	The separation of heterogeneous catalyst from the reaction mixture is easy.
Temperature Dependence	
Homogeneous catalysts work better in low-temperature conditions (less than 250°C).	Heterogeneous catalysts work better in high-temperature conditions (around 250 to 500°C).
Catalyst Modification	
The modification of homogeneous catalysts is easy.	The modification of heterogeneous catalysts is difficult.